Cambridge Assessment International Education

# Cambridge IGCSE™ (9–1)

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| CANDIDATE NAME | |

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*7843989580*

**CHEMISTRY** **0971/41**

Paper 4 Theory (Extended) **May/June 2022**

**1 hour 15 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

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**1** A list of substances is shown.

| | | | | |
|---|---|---|---|---|
| **aluminium oxide** | **carbon dioxide** | **chlorine** | **diamond** | **ethanol** |
| **glucose** | **iron(III) oxide** | **limestone** | **nitrogen** | **oxygen** |

Answer the questions using the list of substances.

Each substance may be used once, more than once or not at all.

State which of the substances:

**(a)** is a reactant in photosynthesis

..... [1]

**(b)** is the main constituent of bauxite

..... [1]

**(c)** are **two** products of fermentation

..... and ..... [2]

**(d)** is used as a fuel

..... [1]

**(e)** is a gas used to convert iron into steel

..... [1]

**(f)** is a greenhouse gas

..... [1]

**(g)** is a gas that is approximately 78% of clean, dry air

..... [1]

**(h)** is a form of carbon.

..... [1]

[Total: 9]

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**2** **(a)** Atoms are made of protons, neutrons and electrons. Atoms of the same element are known as isotopes.

**(i)** Complete the table.

| particle | relative charge | relative mass |
|---|---|---|
| electron | | $\frac{1}{1840}$ |
| neutron | | |
| proton | +1 | |

[2]

**(ii)** $^{24}_{12}Mg$ and $^{25}_{12}Mg$ are isotopes of magnesium.

Complete the table to show the numbers of electrons, neutrons and protons in these isotopes of magnesium.

| isotope | number of electrons | number of neutrons | number of protons |
|---|---|---|---|
| $^{24}_{12}Mg$ | | | |
| $^{25}_{12}Mg$ | | | |

[2]

**(iii)** Explain why magnesium ions have a charge of 2+.

.....

..... [1]

**(b)** $Mg^{2+}$ ions have the electronic structure 2,8.

Give the formula of the following particles which have the same electronic structure as $Mg^{2+}$ ions.

- a cation (positive ion)

.....

- an anion (negative ion)

.....

- an atom

.....

[3]

[Total: 8]

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**3** This question is about sodium and compounds of sodium.

**(a)** **(i)** Describe the bonding in a metallic element such as sodium.

You may include a diagram as part of your answer.

.....

.....

..... [3]

**(ii)** Describe how solid sodium conducts electricity.

..... [1]

**(b)** Some properties of sodium chloride are shown:

- melting point of 801 °C
- non-conductor of electricity when solid
- conductor of electricity when molten
- soluble in water.

**(i)** Name the type of bonding in sodium chloride.

..... [1]

**(ii)** Explain why sodium chloride conducts electricity when molten.

.....

..... [1]

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**(c)** A student determines the concentration of a solution of dilute sulfuric acid, $H_2SO_4$, by titration with aqueous sodium hydroxide, NaOH.

**step 1** 25.0 $cm^3$ of 0.200 mol / $dm^3$ NaOH is transferred into a conical flask.

**step 2** Three drops of methyl orange indicator are added to the conical flask.

**step 3** A burette is filled with $H_2SO_4$.

**step 4** The acid in the burette is added to the conical flask until the indicator changes colour. The volume of acid is recorded. This process is known as titration.

**step 5** The titration is repeated several times until a suitable number of results is obtained.

**(i)** Name the piece of apparatus used to measure exactly 25.0 $cm^3$ of 0.200 mol / $dm^3$ NaOH in **step 1**.

..... [1]

**(ii)** State the colour change of the methyl orange indicator in **step 4**.

from ..... to ..... [1]

**(iii)** State how the student decides that a suitable number of results have been obtained.

.....

..... [1]

**(iv)** 20.0 $cm^3$ of $H_2SO_4$ reacts with 25.0 $cm^3$ of 0.200 mol / $dm^3$ NaOH.

The equation for the reaction is shown.

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

Calculate the concentration of $H_2SO_4$ using the following steps.

- Calculate the number of moles in 25.0 $cm^3$ of 0.200 mol / $dm^3$ NaOH.

..... mol

- Determine the number of moles of $H_2SO_4$ that react with the NaOH.

..... mol

- Calculate the concentration of $H_2SO_4$.

..... mol / $dm^3$
[3]

[Total: 12]

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**4** This question is about compounds of sulfur.

**(a)** Sulfuric acid, $H_2SO_4$, is manufactured using the Contact process. This manufacture involves four stages.

**stage 1** Molten sulfur burns in air to produce sulfur dioxide.

**stage 2** Sulfur dioxide reacts with oxygen to form sulfur trioxide.

**stage 3** Sulfur trioxide combines with concentrated sulfuric acid to form oleum, $H_2S_2O_7$.

**stage 4** Oleum reacts to form concentrated sulfuric acid.

**(i)** Write a chemical equation for the reaction occurring in **stage 1**.

..... [1]

**(ii)** State the essential conditions that are necessary for **stage 2**. Write an equation for the chemical reaction that occurs.

.....

.....

.....

..... [4]

**(iii)** Write a chemical equation for the reaction occurring in **stage 3**.

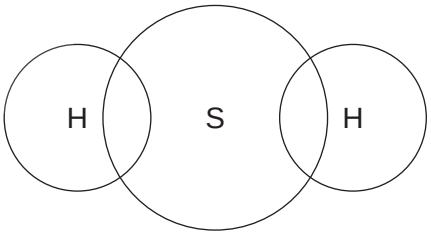
..... [1]

**(iv)** Name the substance that reacts with oleum in **stage 4**.

..... [1]

**(b)** Hydrogen sulfide has the formula $H_2S$.

**(i)** Complete the dot-and-cross diagram to show the electron arrangement in a molecule of hydrogen sulfide. Show outer shell electrons only.




[2]

**(ii)** Balance the chemical equation for the reaction of hydrogen sulfide with sulfur dioxide shown.

$$.....H_2S + SO_2 \rightarrow .....S + .....H_2O$$ [1]

[Total: 10]

**5** Ethanoic acid is manufactured by the reaction of methanol with carbon monoxide.

An equilibrium mixture is produced.

$$CH_3OH(g) + CO(g) \rightleftharpoons CH_3COOH(g)$$

**(a)** State **two** characteristics of an equilibrium.

1 .....

2 .....
[2]

**(b)** The purpose of the industrial process is to produce a high yield of ethanoic acid at a high rate of reaction.

The manufacture is carried out at a temperature of 300 °C.

The forward reaction is exothermic.

Use this information to state why the manufacture is **not** carried out at temperatures:

- **below** 300 °C

.....

- **above** 300 °C.

.....
[2]

**(c)** Complete the table using only the words *increases*, *decreases* or *no change*.

| | effect on the rate of the forward reaction | effect on the equilibrium yield of $CH_3COOH(g)$ |
|---|---|---|
| adding a catalyst | | no change |
| decreasing the pressure | | |

[3]

**(d)** Suggest which of the following metals is a suitable catalyst for the reaction. Give a reason for your answer.

**aluminium** **calcium** **cobalt** **magnesium** **potassium**

suitable catalyst .....

reason .....
[2]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**(e)** Ethanoic acid is a member of the homologous series of carboxylic acids.

State the general formula of this homologous series.

..... [1]

**(f)** Draw the structure of the carboxylic acid containing three carbon atoms. Show all of the atoms and all of the bonds.

[2]

**(g)** When carboxylic acids react with alcohols, esters are produced.

The formula of ester **X** is $CH_3CH_2CH_2COOCH_3$.

**(i)** Name ester **X**.

..... [1]

**(ii)** Give the name of the carboxylic acid and the alcohol that react together to produce ester **X**.

carboxylic acid .....

alcohol .....

[2]

**(h)** Ester **Y** has the following composition by mass:

C, 48.65%; H, 8.11%; O, 43.24%.

Calculate the empirical formula of ester **Y**.

empirical formula = ..... [3]

**(i)** Ester **Z** has the empirical formula $C_2H_4O$ and a relative molecular mass of 88.

Determine the molecular formula of ester **Z**.

molecular formula = ..... [1]

[Total: 19]

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**6** This question is about zinc and its compounds.

**(a)** Zinc is extracted from its ore which is mainly zinc sulfide, ZnS.

The steps for this extraction are shown.

**step 1** Zinc sulfide is converted into zinc oxide.

**step 2** The zinc oxide is then reduced to zinc in a furnace. The zinc formed becomes a gas.

**step 3** The zinc gas is cooled to form molten zinc.

**(i)** Name the ore of zinc, which is mainly zinc sulfide.

..... [1]

**(ii)** Describe how zinc sulfide is converted into zinc oxide in **step 1**.

.....

..... [1]

**(iii)** Name the reducing agent used in **step 2**.

..... [1]

**(iv)** Explain why the zinc forms a gas in **step 2** inside the furnace.

..... [1]

**(v)** State the name of the physical change occurring when zinc gas is converted into molten zinc.

..... [1]

**(b)** Zinc sulfate crystals, $ZnSO_4 \bullet 7H_2O$, are hydrated.

Zinc sulfate crystals are made by reacting zinc carbonate with dilute sulfuric acid.

The equation for the overall process is shown.

$$ZnCO_3 + H_2SO_4 + 6H_2O \rightarrow ZnSO_4 \bullet 7H_2O + CO_2$$

**step 1** Large pieces of solid zinc carbonate are added to dilute sulfuric acid until the zinc carbonate is in excess. This forms aqueous zinc sulfate.

**step 2** The excess zinc carbonate is separated from the aqueous zinc sulfate.

**step 3** The aqueous zinc sulfate is heated until a saturated solution is formed.

**step 4** The saturated solution is allowed to cool and crystallise.

**step 5** The crystals are removed and dried.

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**(i)** In **step 1**, zinc carbonate is in excess when no more zinc carbonate dissolves.

State one **other** observation that indicates the zinc carbonate is in excess in **step 1**.

..... [1]

**(ii)** Name a different substance, other than zinc carbonate, that can be added to dilute sulfuric acid to produce aqueous zinc sulfate in **step 1**.

..... [1]

**(iii)** **Step 1** is repeated using powdered zinc carbonate instead of large pieces.

All other conditions are kept the same.

The rate of reaction increases.

Give a reason why the rate of reaction increases. Explain your answer in terms of particles.

.....

.....

..... [2]

**(iv)** Suggest what is observed when the solution is saturated in **step 3**.

.....

.....

..... [1]

**(v)** The formula of zinc sulfate crystals is $ZnSO_4 \bullet 7H_2O$.

Give the formula of the solid formed if the crystals are heated to dryness in **step 3**.

..... [1]

[Total: 11]

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**7** The Periodic Table can be used to classify elements.

**(a)** Group I elements react with cold water to form alkaline solutions.

**(i)** Place the Group I elements caesium, lithium, potassium, rubidium and sodium in their order of reactivity with water.

Put the most reactive element first.

most reactive → least reactive

| | | | | |
|---|---|---|---|---|
| | | | | |

[1]

**(ii)** Name the alkaline solution formed when caesium reacts with cold water.

..... [1]

**(b)** Group I elements have lower melting points than transition elements.

Describe one **other** difference in the **physical** properties of Group I elements and transition elements.

..... [1]

**(c)** Group VII elements are known as the halogens.

Astatine is below iodine in Group VII.

Predict the physical state of astatine at room temperature and pressure.

..... [1]

**(d)** Some Group VII elements react with aqueous solutions containing halide ions.

When aqueous chlorine is added to aqueous potassium bromide a reaction occurs.

The ionic half-equations for the reaction are shown.

$$Cl_2(aq) + 2e^- \rightarrow 2Cl^-(aq)$$

$$2Br^-(aq) \rightarrow Br_2(aq) + 2e^-$$

**(i)** Describe the colour change of the solution.

original colour of potassium bromide solution .....

final colour of reaction mixture .....

[2]

**(ii)** Identify the species that is oxidised.

Explain your decision.

species oxidised .....

explanation .....
[2]

**(e)** Bromine monochloride, BrC*l*, is made by the reaction between bromine and chlorine. The chemical equation is shown.

$$Br_2(g) + Cl_2(g) \rightarrow 2BrCl(g)$$

| bond | bond energy in kJ / mol |
|---|---|
| Br–Br | 190 |
| C*l*–C*l* | 242 |
| Br–C*l* | 218 |

Calculate the overall energy change for the reaction using bond energies.

Use the following steps.

- Calculate the total amount of energy required to break the bonds in 1 mole of $Br_2(g)$ and 1 mole of $Cl_2(g)$.

..... kJ

- Calculate the total amount of energy released when the bonds in 2 moles of BrC*l*(g) are formed.

..... kJ

- Calculate the overall energy change for the reaction.

..... kJ / mol
[3]

[Total: 11]

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## The Periodic Table of Elements

**Key**

| atomic number |
|---|
| **atomic symbol** |
| name |
| relative atomic mass |

| Group I | Group II | | | | | | | | | | | Group III | Group IV | Group V | Group VI | Group VII | Group VIII |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1<br>**H**<br>hydrogen<br>1 | | | | | | | | | | 2<br>**He**<br>helium<br>4 |
| 3<br>**Li**<br>lithium<br>7 | 4<br>**Be**<br>beryllium<br>9 | | | | | | | | | | | 5<br>**B**<br>boron<br>11 | 6<br>**C**<br>carbon<br>12 | 7<br>**N**<br>nitrogen<br>14 | 8<br>**O**<br>oxygen<br>16 | 9<br>**F**<br>fluorine<br>19 | 10<br>**Ne**<br>neon<br>20 |
| 11<br>**Na**<br>sodium<br>23 | 12<br>**Mg**<br>magnesium<br>24 | | | | | | | | | | | 13<br>**Al**<br>aluminium<br>27 | 14<br>**Si**<br>silicon<br>28 | 15<br>**P**<br>phosphorus<br>31 | 16<br>**S**<br>sulfur<br>32 | 17<br>**Cl**<br>chlorine<br>35.5 | 18<br>**Ar**<br>argon<br>40 |
| 19<br>**K**<br>potassium<br>39 | 20<br>**Ca**<br>calcium<br>40 | 21<br>**Sc**<br>scandium<br>45 | 22<br>**Ti**<br>titanium<br>48 | 23<br>**V**<br>vanadium<br>51 | 24<br>**Cr**<br>chromium<br>52 | 25<br>**Mn**<br>manganese<br>55 | 26<br>**Fe**<br>iron<br>56 | 27<br>**Co**<br>cobalt<br>59 | 28<br>**Ni**<br>nickel<br>59 | 29<br>**Cu**<br>copper<br>64 | 30<br>**Zn**<br>zinc<br>65 | 31<br>**Ga**<br>gallium<br>70 | 32<br>**Ge**<br>germanium<br>73 | 33<br>**As**<br>arsenic<br>75 | 34<br>**Se**<br>selenium<br>79 | 35<br>**Br**<br>bromine<br>80 | 36<br>**Kr**<br>krypton<br>84 |
| 37<br>**Rb**<br>rubidium<br>85 | 38<br>**Sr**<br>strontium<br>88 | 39<br>**Y**<br>yttrium<br>89 | 40<br>**Zr**<br>zirconium<br>91 | 41<br>**Nb**<br>niobium<br>93 | 42<br>**Mo**<br>molybdenum<br>96 | 43<br>**Tc**<br>technetium<br>– | 44<br>**Ru**<br>ruthenium<br>101 | 45<br>**Rh**<br>rhodium<br>103 | 46<br>**Pd**<br>palladium<br>106 | 47<br>**Ag**<br>silver<br>108 | 48<br>**Cd**<br>cadmium<br>112 | 49<br>**In**<br>indium<br>115 | 50<br>**Sn**<br>tin<br>119 | 51<br>**Sb**<br>antimony<br>122 | 52<br>**Te**<br>tellurium<br>128 | 53<br>**I**<br>iodine<br>127 | 54<br>**Xe**<br>xenon<br>131 |
| 55<br>**Cs**<br>caesium<br>133 | 56<br>**Ba**<br>barium<br>137 | 57–71<br>lanthanoids | 72<br>**Hf**<br>hafnium<br>178 | 73<br>**Ta**<br>tantalum<br>181 | 74<br>**W**<br>tungsten<br>184 | 75<br>**Re**<br>rhenium<br>186 | 76<br>**Os**<br>osmium<br>190 | 77<br>**Ir**<br>iridium<br>192 | 78<br>**Pt**<br>platinum<br>195 | 79<br>**Au**<br>gold<br>197 | 80<br>**Hg**<br>mercury<br>201 | 81<br>**Tl**<br>thallium<br>204 | 82<br>**Pb**<br>lead<br>207 | 83<br>**Bi**<br>bismuth<br>209 | 84<br>**Po**<br>polonium<br>– | 85<br>**At**<br>astatine<br>– | 86<br>**Rn**<br>radon<br>– |
| 87<br>**Fr**<br>francium<br>– | 88<br>**Ra**<br>radium<br>– | 89–103<br>actinoids | 104<br>**Rf**<br>rutherfordium<br>– | 105<br>**Db**<br>dubnium<br>– | 106<br>**Sg**<br>seaborgium<br>– | 107<br>**Bh**<br>bohrium<br>– | 108<br>**Hs**<br>hassium<br>– | 109<br>**Mt**<br>meitnerium<br>– | 110<br>**Ds**<br>darmstadtium<br>– | 111<br>**Rg**<br>roentgenium<br>– | 112<br>**Cn**<br>copernicium<br>– | | 114<br>**Fl**<br>flerovium<br>– | | 116<br>**Lv**<br>livermorium<br>– | | |

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| lanthanoids | 57<br>**La**<br>lanthanum<br>139 | 58<br>**Ce**<br>cerium<br>140 | 59<br>**Pr**<br>praseodymium<br>141 | 60<br>**Nd**<br>neodymium<br>144 | 61<br>**Pm**<br>promethium<br>– | 62<br>**Sm**<br>samarium<br>150 | 63<br>**Eu**<br>europium<br>152 | 64<br>**Gd**<br>gadolinium<br>157 | 65<br>**Tb**<br>terbium<br>159 | 66<br>**Dy**<br>dysprosium<br>163 | 67<br>**Ho**<br>holmium<br>165 | 68<br>**Er**<br>erbium<br>167 | 69<br>**Tm**<br>thulium<br>169 | 70<br>**Yb**<br>ytterbium<br>173 | 71<br>**Lu**<br>lutetium<br>175 |
| actinoids | 89<br>**Ac**<br>actinium<br>– | 90<br>**Th**<br>thorium<br>232 | 91<br>**Pa**<br>protactinium<br>231 | 92<br>**U**<br>uranium<br>238 | 93<br>**Np**<br>neptunium<br>– | 94<br>**Pu**<br>plutonium<br>– | 95<br>**Am**<br>americium<br>– | 96<br>**Cm**<br>curium<br>– | 97<br>**Bk**<br>berkelium<br>– | 98<br>**Cf**<br>californium<br>– | 99<br>**Es**<br>einsteinium<br>– | 100<br>**Fm**<br>fermium<br>– | 101<br>**Md**<br>mendelevium<br>– | 102<br>**No**<br>nobelium<br>– | 103<br>**Lr**<br>lawrencium<br>– |

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).

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